

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – College Admission Agent (RAG Based)

Domain of Project – Education

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Problem Statement -

Problem Statement : College Admission Agent

The Challenge

Choosing the right college is difficult because students must compare **institute brochures, counselling rules, cutoff trends, fee structures, and eligibility criteria** across many sources. This information is often scattered, inconsistent, and difficult to interpret. Without **personalized, data-backed guidance**, students may miss suitable colleges, misunderstand admission chances, or make decisions without knowing the actual fee and cutoff details.

The Objective

Build an **AI-powered College Admission Agent** that provides **personalized admission insights and guidance** for students, parents, and counsellors. The solution should:

- **Personalized College Shortlisting** – Recommend colleges based on rank, marks, category, stream, location, budget, and preferences.
- **Admission Data RAG Retrieval** – Fetch and summarize cutoff, fee, eligibility, and counselling details from trusted brochures and documents.
- **College Comparison & Guidance** – Compare institutes across branch availability, fees, placements, and admission chances.
- **Counselling Support** – Assist with round-wise cutoff trends, seat availability, and admission strategy.
- **Multi-Agent System** –
 - **Institute Knowledge Agent** (retrieves brochure and admission data)
 - **College Recommendation Agent** (shortlists suitable institutes)
 - **Admission Advisory Agent** (explains eligibility and counselling options)
 - **Document Query Agent** (answers brochure-based questions)
- **Visualization Dashboard** – Show shortlist, cutoff comparison, fee comparison, and admission probability insights.

Technology Used

Technology Used

- Tech Stack
- IBM watsonx Orchestrate
- IBM watsonx.ai
- IBM Granite Models
- RAG (Retrieval-Augmented Generation)
- Vector Database
- PDF Knowledge Base

Knowledge Sources

- JoSAA Bulletin
- CSAB Bulletin
- JAC Delhi
- JAC Chandigarh
- UPTAC 2026 Bulletin
- Other official Counselling PDFs

Why This Stack?

watsonx Orchestrate → To manage and automate the agent workflow

watsonx.ai → To power AI reasoning and response generation

Granite Models → For summarization, analysis, and intelligent answers

RAG → To retrieve accurate information from official admission PDFs

Vector Database → To store and search document embeddings efficiently

Proposed Solution -

Proposed Solution: AI-Powered College Admission Agent

The proposed solution is a **multi-agent AI-powered College Admission Agent** built using **IBM Orchestrate**, **IBM watsonx.ai**, and **IBM Granite models** to provide personalized, reliable admission guidance. The system addresses the challenge of college selection, cutoff analysis, and fee comparison through an intelligent modular architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch accurate admission data from trusted sources such as **college brochures, counselling PDFs, cutoff lists, and fee documents**. This data is embedded and stored in a vector database, enabling the system to retrieve and summarize college details, admission criteria, and fee structures using Granite models.

The system consists of four key agents:

Institute Knowledge Agent retrieves and summarizes brochure-based admission information using RAG.

College Recommendation Agent generates personalized college shortlists based on rank, category, budget, branch, and location.

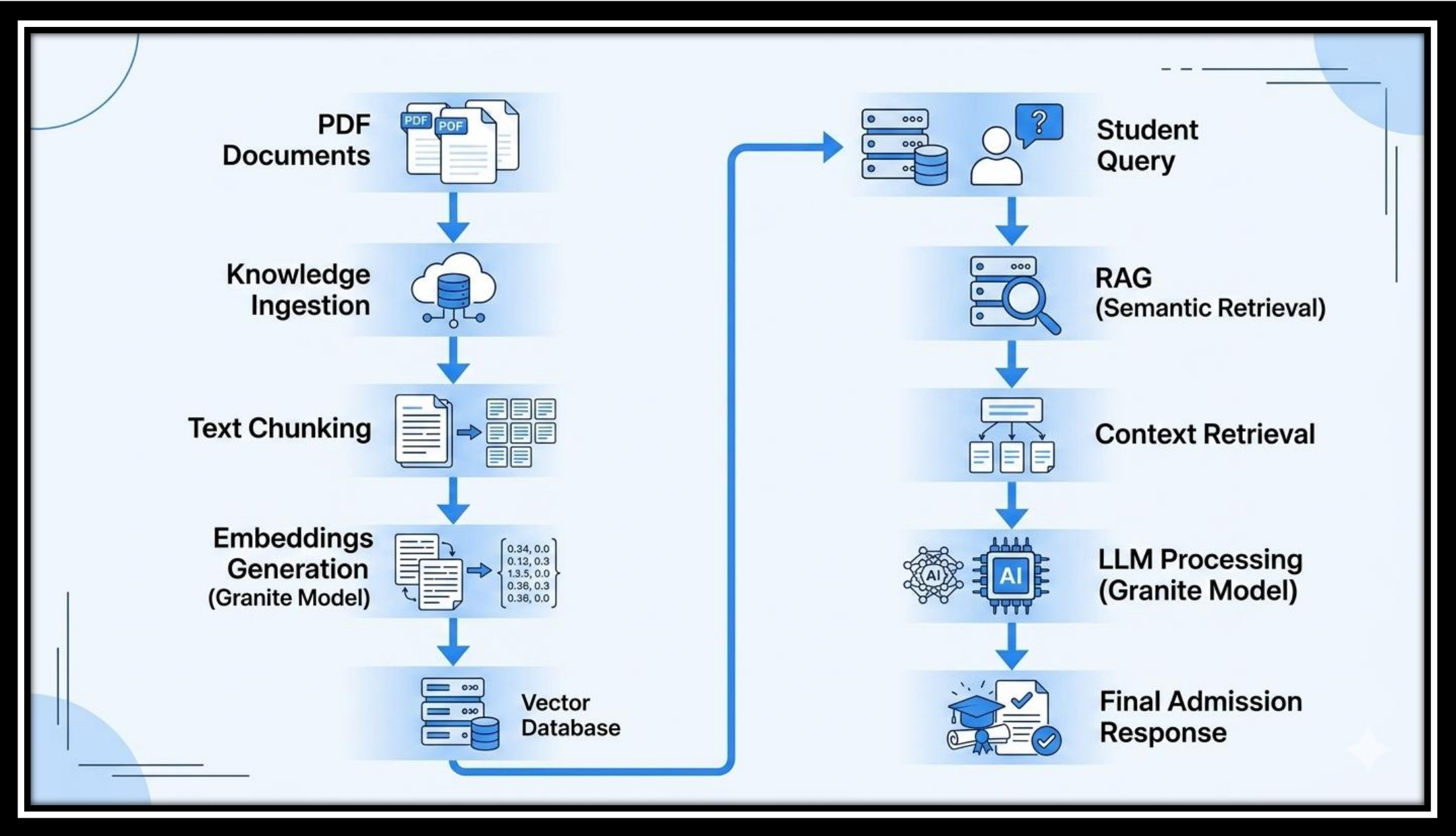
Admission Advisory Agent provides counselling guidance, cutoff interpretation, and eligibility support.

Document Query Agent allows users to ask questions from uploaded brochures and admission documents and delivers instant answers.

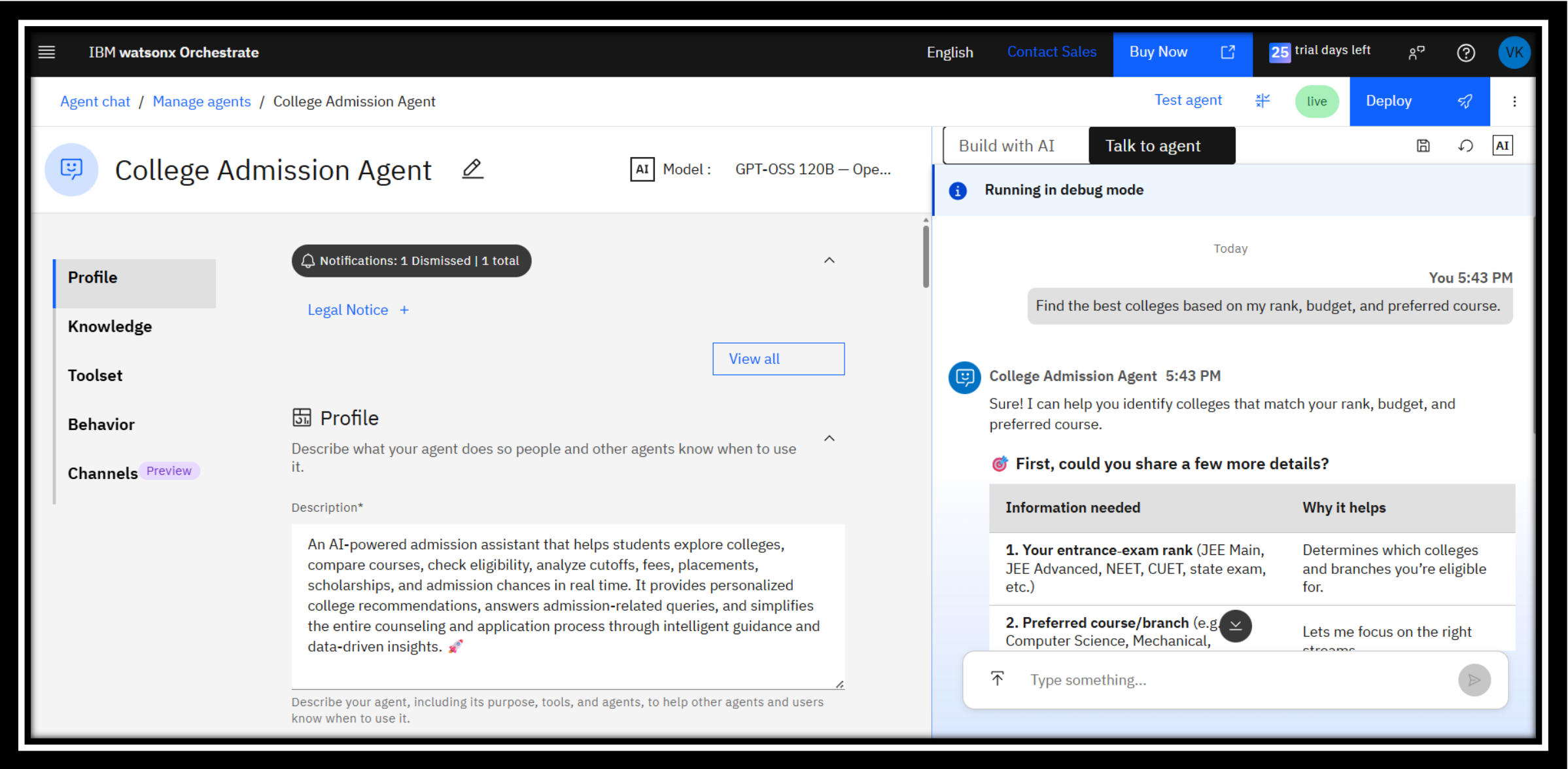
IBM Orchestrate is used to coordinate agent workflows, while **watsonx.ai** ensures scalable model deployment, embeddings, and AI governance. **Granite models** power reasoning, summarization, and personalized admission recommendations.

Overall, this solution delivers an **end-to-end intelligent admission assistant** that helps students make better college choices through personalized insights, document-grounded answers, and counselling support.

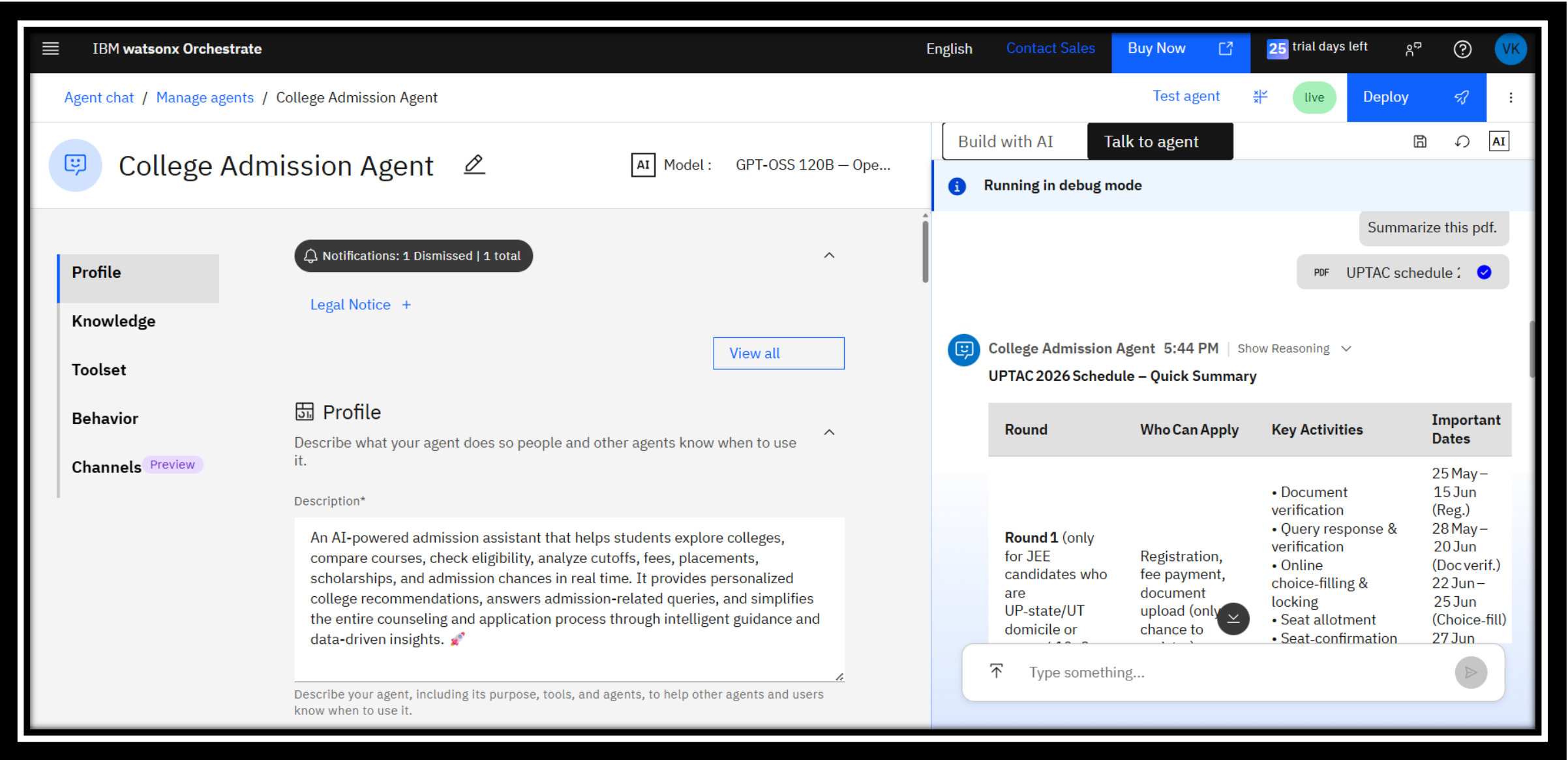
IBM Orchestrate workflow -



Project image



Project image



Role of Agentic AI in the solution

Role of Agentic AI in College Admission Agent

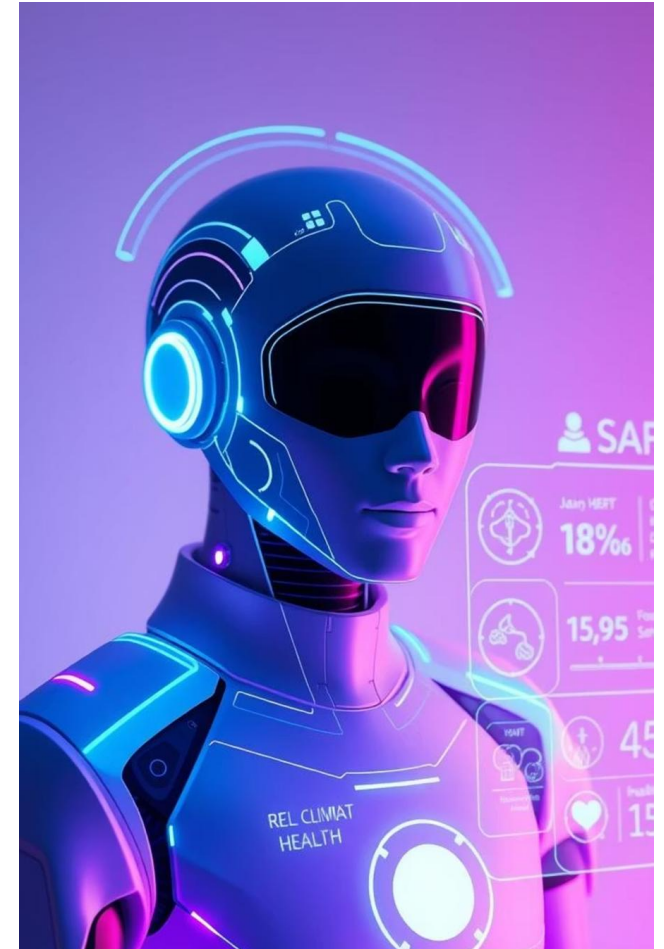
Agentic AI enables the College Admission Agent to function as an intelligent, autonomous system rather than a simple chatbot. Using **IBM watsonx.ai**, **IBM Orchestrate**, and **IBM Granite models**, it coordinates multiple specialized agents to deliver personalized admission guidance.

Each agent performs a specific role—such as retrieving brochure data, analyzing cutoff trends, comparing fees, and recommending colleges—while working collaboratively. This **multi-agent approach** ensures accurate shortlist generation and better admission decisions.

Agentic AI also brings **context awareness**, understanding user details like rank, category, preferred branch, budget, and location to generate tailored recommendations. It supports **real-time adaptation**, continuously updating suggestions based on new counselling rounds, uploaded brochures, and user inputs.

Additionally, agents communicate with each other, enabling better comparison, reasoning, and decision-making.

Overall, Agentic AI makes the system proactive, personalized, and admission-focused, acting like a **smart digital admission counsellor** that helps users choose the right college with confidence.



Novelty and Uniqueness

The proposed College Admission Agent stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent admission system.

Multi-Agent Intelligence – Instead of a single chatbot, multiple specialized agents collaborate to deliver accurate and context-aware admission guidance.

RAG-Based Accuracy – Integrates brochure and admission document retrieval with AI generation, reducing misinformation and ensuring data-backed answers.

Hyper-Personalization – Tailors college recommendations based on rank, category, branch, budget, location, and counselling preferences.

Admission Comparison Loop – Continuously adapts shortlists based on cutoff trends, fee data, and user feedback.

Document-Centric Input – Supports brochure, PDF, and text-based admission data for reliable question answering.

Counselling Support Focus – Goes beyond basic search to provide branch-wise cutoff insights and admission strategy guidance.

Scalable & Enterprise-Ready – Built on **IBM Orchestrate** with **IBM Granite models**, ensuring reliability, governance, and modular workflow management.

This makes the solution a **smart, adaptive, and data-grounded admission assistant**, not just a static college search tool.

Git Hub Link

Repository Details

A public GitHub repository has been created to host all project resources, source files, and documentation related to the **College Admission Agent (RAG Based)** project.

Repository Contents

- ✓ Problem Statement PDF
- ✓ Project Presentation (PPTX)
- ✓ Knowledge Base Documents (JoSAA, CSAB, JAC Delhi, JAC Chandigarh, UPTAC, etc.)
- ✓ Additional Project Files and Resources

GitHub Repository URL

 <https://github.com/codecraftpython-dev/College-Admission-Agent>

Future Scope

1. Integration with Counselling Portals & Seat Matrix

The system can be integrated with counselling portals, seat matrix updates, and deadline alerts to collect real-time admission information such as round-wise cutoffs, vacant seats, and reporting schedules. This will enable the College Admission Agent to provide more accurate and dynamic recommendations based on actual counselling progress and institute-wise changes.

2. AI-Powered Voice Assistant & Multilingual Support

Enhancing the system with a voice-enabled assistant and support for multiple languages will improve accessibility. Students can interact naturally in their preferred language, making the solution more inclusive and suitable for diverse users, especially during admission seasons where quick guidance is needed.

Thank You!

Thank you for your time and interest.